Gating AI Deployment on Environmental Evidence Quality

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Abstract

Carbon-aware governance proposals gate AI deployment on impact magnitude: a carbon budget, a readiness level, or a risk rating. None of them asks whether the carbon number itself can be trusted. We present FairMind-E, an environmental governance module deployed inside FairMind, an open-source AI governance platform, that makes measurement provenance a first-class release criterion: a provenance-derived confidence score crossed with an impact tier resolves to *go*, *conditional_go*, or *no_go*, and blocks the platform's deployment-approval workflow. In a measurement study over four workloads, five iterations each, every physical run was assessed by a software power tracker and an analytic FLOP model. The two disagreed on magnitude by 123–169 percent MAPE depending on which is taken as reference, and because the policy assigns confidence by provenance class, that disagreement placed them in different columns of the matrix on all 20 paired runs while the impact tier stayed *low* throughout. We are explicit that this split is a consequence of the policy rather than an independent confirmation of it; what the runs establish empirically is the size of the method disagreement and the behaviour under mitigation. There a routine claim failed outright: casting a fine-tune to `bfloat16` on a host with no optimised kernel made it 46 times slower, and the tracker estimated roughly 41 times the energy, so "we reduced precision, therefore we cut carbon" was false in the harmful direction. Evidence quality, not only impact magnitude, decides whether an environmental claim can carry a release.

1 The gap: a measurement layer with no trust layer

Regulators and standards bodies are building the measurement layer for AI's environmental footprint: quantification-methods inventories [1], sustainability overviews [2], and a G7-commissioned inventory cataloguing thirteen measurement, monitoring, and reporting initiatives [3]. The EU AI Act anticipates harmonised standards for "reporting and documentation processes to improve AI systems' resource performance" (Article 40), and general-purpose AI documentation must state known or estimated energy consumption (Annex XI) [4]. Yet none of these instruments defines how the *quality* of an environmental claim should condition a deployment decision.

Recent governance research gates on magnitude. Carbon-aware governance gates embed carbon budgets in governance layers, recording provenance only in audit logs [5]. Sustainability-pyramid frameworks gate on readiness levels [6], and risk-management integrations treat AI carbon as a risk dimension in banking frameworks [7]. Meanwhile measurement research has established that reliability varies sharply with disclosure: metadata-completeness tiers across 5,227 public models span roughly $1.2\times$ to $2\times$ estimation error [8], and spend-based fallbacks in tiered Scope 3 accounting overestimate by 10–40$\times$ [9]. Adjacent work has begun to act on estimate trust, through conformal calibration inside a carbon budget gate [10] and confidence-gated filtering of industrial emissions

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predictions [11]. Those act on the statistical spread of an estimate: conformal calibration can block a wide-interval prediction. None acts on where the number came from, so a precise-looking figure from a weak source, a narrow interval on a vendor-reported or analytic estimate, passes every published gate while sitting comfortably under budget.

Our claim is narrow: environmental *evidence quality*, not only impact magnitude, should gate AI deployment. The ancestry is the data-quality tradition in life-cycle assessment, where pedigree matrices grade the provenance of an inventory figure [12], and the NUSAP uncertainty pedigree [13]; recent work recommends indicator-style disclosure for AI-assisted footprinting but proposes no score, thresholds, or gate [14]. We invert it: where that work assesses AI producing carbon numbers, we score carbon numbers to gate the AI.

2 Mechanism

Provenance classes and confidence. Every assessment declares its measurement source, normalized to one of five provenance classes with bounded confidence ranges: *measured* (hardware telemetry; 0.85–1.00), *tool_estimated* (software power trackers; 0.60–0.85), *vendor_reported* (cloud-provider figures; 0.40–0.60), *manual* (analytic FLOP estimates; 0.20–0.40), and *unknown* (0.0). A caller-supplied score is honoured but never allowed to exceed its provenance cap, and unknown provenance forces 0.0 regardless of any claimed score. Provenance is categorical and reported separately from quantified uncertainty, so a precise-looking number cannot launder a weak source.

Tiers and the decision matrix. Metrics map to low, medium, or high impact per lifecycle phase, using location-based carbon so market instruments cannot lower a gate outcome. Confidence bands, named to stay distinct from the provenance classes above (*high* ≥ 0.70; *moderate* $0.35 \leq c < 0.70$; *low* $0 < c < 0.35$; *undisclosed* $c = 0$), cross with impact tiers into the matrix shown in Figure 1(b). Confidence 0.0 short-circuits to *no_go* before matrix lookup: an undisclosed claim never passes. A *conditional_go* cannot be approved until it carries a dated, owned mitigation or an explicit exception with owner, expiry, and rationale, which makes reviewer overrides auditable rather than informal. Six controls (ENV-1 to ENV-6: boundary, energy accounting, dual location and market carbon, per-functional-unit efficiency, evidence confidence with disclosed uncertainty, and mitigation or exception) define coverage. Disclosed offsets are recorded but never subtracted from emissions and never improve any gate output.

3 Implementation

FairMind-E ships inside FairMind, an open-source AI governance platform, not as a standalone proposal. The decision engine is a framework-agnostic Python package with 57 unit tests covering every matrix cell, band boundary, provenance cap, and the undisclosed-evidence short-circuit. Assessments persist as versioned governance evidence, and a release gate hooks the platform's deployment-approval workflow, returning HTTP 409 with the blocking reason when the latest assessment is *no_go* or an unmitigated *conditional_go*. Reviewers see the matrix position, control coverage, mitigations, and version trail; disclosures export in a form shaped to the European climate reporting standard [15].

A 14-case hand-labelled fixture set spanning all matrix cells, mitigation states, exception paths, and offset disclosures serves as a *conformance test*: the implementation reproduces 14/14 of the specified labels, and ablations that drop the confidence dimension reproduce far fewer. Since the authors wrote the fixtures, the labels, the engine, and the ablations, this establishes only that the code implements the stated policy. It is not evidence that the policy decides correctly, and we do not report it as a baseline comparison.

4 Measurement study: what the runs show, and what they cannot

Four workloads (classical ML training, transformer fine-tune, RAG inference, batch embeddings) ran five iterations each on a single host. Every physical run was assessed by three evidence paths: a software power tracker (*tool_estimated*), operator-filled cloud billing (*vendor_reported*), and a documented analytic FLOP model (*manual*). Only two yielded figures. The harness never fabricates:

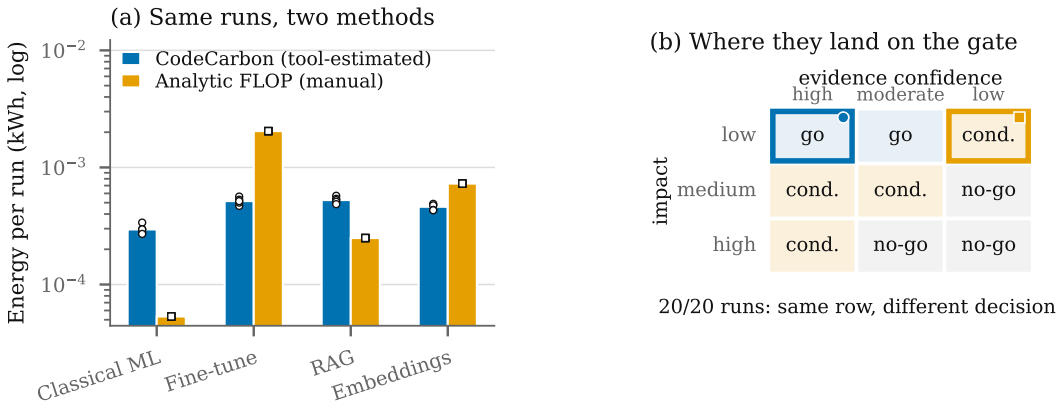


Figure 1: Four workloads, five iterations each, every run assessed twice. (a) The empirical result: the methods disagree by 123–169 percent MAPE depending on reference; points are iterations, bars their means. (b) Where those runs land. Both put every run in the same *low* row, so the column moved because provenance moved. With confidence assigned per provenance class, the differing outcome follows from the policy once both sit in one tier: the panel illustrates the mechanism rather than validating it.

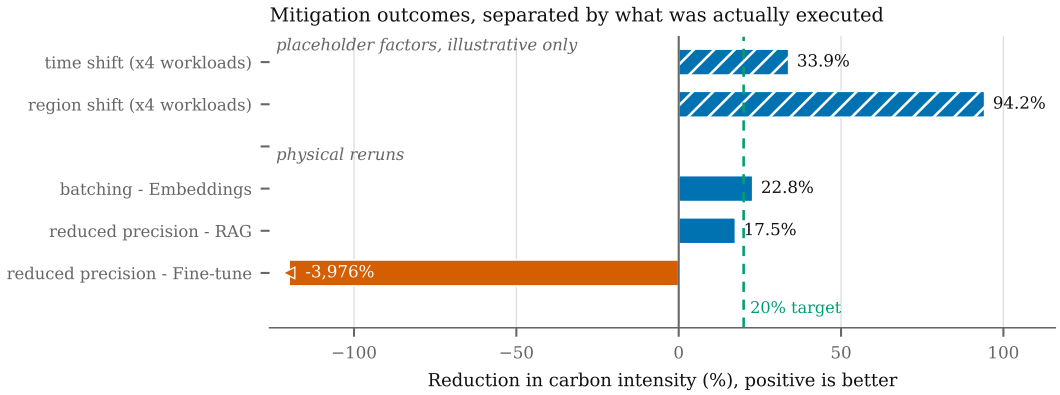


Figure 2: Mitigation outcomes, separated by what was executed. Hatched bars are accounting scenarios: one measured run recomputed against a bundled *placeholder* emission-factor table with no rerun, so they are illustrative arithmetic, not measurements, hence identical across workloads and collapsed to one bar each. Solid bars are the three physical reruns. Casting the fine-tune to `bfloat16` on a host with no optimised kernel took it from 62.7 s to 2887.6 s; its bar is truncated.

cloud billing lags execution by hours to days, so its 20 records are explicit unavailable entries with a stated reason. All runs were replayed through the production engine, not a reimplementation.

The empirical finding is the disagreement. MAPE is 122.6 percent with the tracker as reference and 169.1 percent with the FLOP model as reference, from 52 to 298 percent by workload, with Bland-Altman limits of agreement spanning -1.15×10^{-3} to 1.79×10^{-3} kWh (Figure 1(a)). Two methods a practitioner might reasonably choose disagree by more than a factor of two on the same physical run.

The decision split is a policy consequence, stated as such. The tracker maps to *tool_estimated*, confidence 0.72, *high* band, *go*; the FLOP model to *manual*, 0.30, *low* band, *conditional_go*, on all 20 runs. Confidence is a constant per provenance class and the tier was *low* throughout, so once both sit in one tier the matrix entails that the outcomes differ: the count follows from the policy and is not independent evidence for it. What it does establish is that the mechanism runs end to end on real measurements, and that a magnitude-only gate is structurally blind to this split. Whether the hierarchy is *correct* needs calibration against an independent power reference: the essential next experiment.

Mitigations: three reruns and eight accounting scenarios. We pre-specified a paired test that mitigations cut intensity by more than 20 percent. We do not report it, because the pairs are not exchangeable replications (Figure 2): only three are physical reruns, while the region and time shifts recompute one measured run against a bundled placeholder emission-factor table with no rerun at all, which is why they are identical across workloads. Pooling accounting scenarios with physical reruns, and different interventions with each other, would give a fictitious degrees of freedom; we report the classes separately instead.

The region- and time-shift scenarios are illustrative arithmetic on placeholder emission factors, identical by construction across workloads; they are not measurements and we report no reduction figure for them. Of the three physical reruns, batching cut intensity 22.8 percent and a reduced-precision cast on RAG 17.5 percent. The third inverted: casting the fine-tune to `bfloat16` made it 46 times slower on a host with no optimised kernel, and the tracker estimated roughly 41 times the energy. The runtime regression is directly measured; the energy figure inherits the tracker's power model.

This is the mitigation analogue of the paper's argument. "We reduced precision, therefore we cut carbon" is routine and entirely plausible; here it was false in the harmful direction. A gate that accepts asserted mitigations would have logged a reduction that never happened, which is why ENV-6 requires the mitigation to be documented and re-measured rather than declared.

Expert panel (designed, collection not started). Twelve engine-validated profiles covering all nine matrix cells are to be rated by 15 reviewers across governance, ML engineering, and sustainability roles under a two-arm counterbalanced design (baseline: raw measurements only; module: full FairMind-E output), testing whether displayed confidence correlates with conservatism, whether the module arm raises ENV control coverage, and whether the matrix improves inter-rater agreement. Instruments and analysis are implemented, recruitment has not begun, and pre-registration is planned before collection; we seek critique of the baseline arm before collection opens.

5 Pathway to climate impact

Environmental reporting for AI is moving from voluntary to mandatory, but mandatory reporting only reduces emissions if the reported numbers are trustworthy enough to act on. Our measurement study shows the gap is not hypothetical: two standard methods applied to the same run disagreed by more than a factor of two, and a textbook efficiency intervention made a workload 46 times slower while a declaration-based gate would have logged it as a reduction. Making evidence quality a release criterion changes the incentive directly: a team that cannot substantiate its number cannot ship unconditionally, so measurement effort becomes the cheapest path to approval rather than an unrewarded cost. This is a leverage point, not abatement: it does not cut emissions, it makes reported reductions verifiable and therefore actionable. The near-term channel is regulatory, since Article 40's harmonised standards do not yet exist [4, 16].

Limitations. The host exposes no hardware energy counters, so the tracker fell back to load-based estimation: this study compares an estimate against an estimate and never claims the *measured* class. These are benchmark workloads, so only the low-impact row is exercised and the provisional thresholds cannot be recalibrated here. The FLOP model uses placeholder constants, and confidence is not calibrated against an independent power reference. Code, run data and analysis scripts are public (Appendix A).

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Appendix A: Reproducibility

Supplementary; read at the reviewers' discretion. Every value below is taken from the committed study artifacts rather than restated from memory.

Host and software. Apple M1 Pro, arm64, macOS. Python 3.13, scikit-learn 1.9.0, numpy 2.4.6, torch 2.13.0, codecarbon 3.2.9. Mains power, other applications closed, warm-up iteration discarded.

Tracker configuration and fallback. `OfflineEmissionsTracker` with a country ISO code, sampling every 15 s in real mode. On this host codecarbon logs *No CPU tracking mode found. Falling back on CPU load mode*: Apple Silicon exposes no RAPL counters and `powermetrics` requires root, so its figures are a CPU-load model against a TDP constant, not a hardware reading. This is why the study never claims the *measured* provenance class.

Workloads. Five iterations each, identical seed (1337) across iterations so spread isolates measurement noise rather than data variation. Configurations were scaled so each run spans several tracker sampling windows; at default sizes runs finished in 2.0–60.5 s, below one window.

Workload	Functional unit	Scaled configuration
Classical ML training	1000 training rows	100k samples, 40 features, 5-fold, 200 iter, 120 repeats
Transformer fine-tune	1000 examples	d_{model} 128, 2 layers, 4 heads, seq 128, batch 16, 2600 steps
RAG inference	1000 queries	200k docs, d 256, 5000 queries, batch 256, top-10, 7 repeats
Batch embeddings	10000 embeddings	500k items, 384/512/256, batch 1024, 32 repeats

Analytic FLOP model. Per workload, with *placeholder* hardware constants of 4.0 GFLOP/s per watt and PUE 1.0: classical ML $=$ repeats $\cdot$ folds $\cdot\ 2 \cdot (n \cdot \frac{\text{folds}-1}{\text{folds}}) \cdot d \cdot$ max_iter; RAG $=$ repeats $\cdot\ (2n_q d_{in} d_{emb} + 2n_q n_{docs} d_{emb})$; batch embeddings $=$ repeats $\cdot\ (2nd_{in}h + 2nhd_{out})$. Documented simplifications (early solver convergence, activations, top-k selection) are recorded in each workload module.

Impact-tier thresholds (provisional). Training: low < 100, high $> 10^4$ kgCO2e. Fine-tune: low < 50, high > 5000 kgCO2e. Inference: low $< 10^{-3}$, high $> 10^{-2}$ kgCO2e per 1000 requests. LLM inference: low < 1, high > 10 kgCO2e per 1M tokens. Tiering uses location-based carbon. These remain provisional: benchmark-scale runs sit orders of magnitude below the low ceilings and cannot recalibrate them.

Grid intensity. Bundled placeholder table, 2023 edition, e.g. IND 713 gCO2e/kWh location and market; SWE 41/30. Region and time-shift figures are recomputations against this table, not reruns, and must be replaced with an authoritative versioned source before any quantitative claim rests on them.

Reproduce. From the repository root, with the backend environment: run the harness (`-workload all -iterations 5 -seed 1337` with the archived override file), then replay every row through the production engine, then compute the agreement statistics. Functional units for the phase-tiered inference workloads are derived from each workload's configuration with the same overrides applied, so the replay reproduces without a hand-written mapping. The harness, the override file (`measurement/overrides_m1.json`), the raw run data (`results/measurement/`) and the analysis scripts are in `research/fairmind-e` of the FairMind repository, `https://github.com/adhit-r/fairmind`, archived as release v0.3.0-alpha at doi:10.5281/zenodo.24043235.